

Departement Elektriese, Elektroniese en  
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and  
Computer Engineering

## Eksamenvraestel

Kopiereg voorbehou

## Examination Question Paper

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### Module EBN122

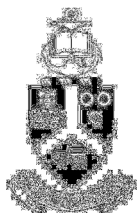
Elektrisiteit en Elektronika

19 November 2012

### Module EBN122

Electricity and Electronics

19 November 2012



UNIVERSITEIT VAN PRETORIA  
UNIVERSITY OF PRETORIA  
YUNIBESITHI YA PRETORIA

#### Eksamensinligting:

#### Exam information:

|                                                                                                        |                                  |                                                  |                      |
|--------------------------------------------------------------------------------------------------------|----------------------------------|--------------------------------------------------|----------------------|
| Maksimum punte:<br><i>Maximum marks:</i>                                                               | 90                               | Volpunte:<br><i>Full marks:</i>                  | 90                   |
| Duur van vraestel:<br><i>Duration of paper:</i>                                                        | 180 minute<br><i>180 minutes</i> | Oopboek / toeboek:<br><i>Open / closed book:</i> | Toe<br><i>Closed</i> |
| Totale aantal bladsye (hierdie blad ingesluit):<br><i>Total number of pages (including this page):</i> |                                  | 28                                               |                      |

#### BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.  
*The examination regulations of the University of Pretoria apply.*
- Alle vrae moet op die ANTWOORDBLAD beantwoord word.  
*All questions must be answered on the ANSWER SHEET.*
- Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.  
*Students may do their own rough calculations on the question paper – the question paper will not be taken in.*
- Antwoorde moet **eenhede** insluit!  
*Answers must include **units**!*
- Finale antwoorde moet as **reële getalle** geskryf word, en **nie** as **breuke** nie.  
*Final answers must be written as **real numbers**, and **not** as **fractions**.*
- Polêre vorm:  $A \angle \theta$  met  $A > 0$ ; Reghoekige vorm:  $a + jb$   
*Polar form:  $A \angle \theta$  with  $A > 0$ ; Rectangular form:  $a + jb$*

|                                                     |                                        |                                                    |                     |
|-----------------------------------------------------|----------------------------------------|----------------------------------------------------|---------------------|
| Interne Eksaminatore:<br><i>Internal Examiners:</i> | Prof. J. Joubert<br>Mnr/Mr. D. le Roux | Eksterne Eksaminator:<br><i>External Examiner:</i> | Prof. J.W. Odendaal |
|-----------------------------------------------------|----------------------------------------|----------------------------------------------------|---------------------|

## **VRAAG/QUESTION 1 [20]**

### **Vraag 1.1 [4]**

- (a) Die lading wat by die positiewe terminal van 'n element ingaan is  $q(t) = 10 \sin(4\pi t)$  mC, terwyl die spanning oor die element gegee word deur  $v(t) = 2\cos(4\pi t)$  V. Bereken die drywing gelewer aan die element by  $t = 0.3$ s.
- (b) Die stroom wat by die positiewe terminal van 'n element ingaan is  $i(t) = 10e^{-2t}$  mA, terwyl die spanning oor die element gegee word deur  $v(t) = 5e^{-2t}$  V. Bereken die energie wat aan die element gelewer word tussen 0 en 0.5s.

### **Question 1.1 [4]**

- (a) The charge entering the positive terminal of an element is  $q(t) = 10 \sin(4\pi t)$  mC, while the voltage across the element is given by  $v(t) = 2\cos(4\pi t)$  V. Find the power delivered to the element at  $t = 0.3$ s.
- (b) The current entering the positive terminal of an element is  $i(t) = 10e^{-2t}$  A, while the voltage across the element is given by  $v(t) = 5e^{-2t}$  V. Calculate the energy delivered to the element between 0 and 0.5s.

**Vraag 1.2 [4]**

- (a) Indien 'n stroom van 200 mA deur 'n weerstand vloei en 40 J elektriese energie word in 30s omgesit na hitte, bepaal die spanningsval oor die weerstand.
- (b) Die vervaardiger van 'n 6V battery verklaar dat die battery 15 mA for 60 ure aanmekeer kan lewer. Gedurende hierdie tyd sal die spanning val van 6V tot 4V. Aanvaar dat die spanningsval lineêr met tyd is, en bereken die hoeveelheid energie wat deur die battery in die 60 ure gelewer word.

**Question 1.2 [4]**

- (a) If a current of 200 mA passes through a resistor and 40 J of electrical energy is converted to heat in 30s, determine the potential drop across the resistor.
- (b) The manufacturer of a 6V battery says that the battery will deliver 15mA for 60 continuous hours. During that time the voltage will drop from 6V to 4V. Assume the voltage is linear with time, and calculate how much energy is delivered by the battery in this 60 hour interval.

**Vraag 1.3 [4]**

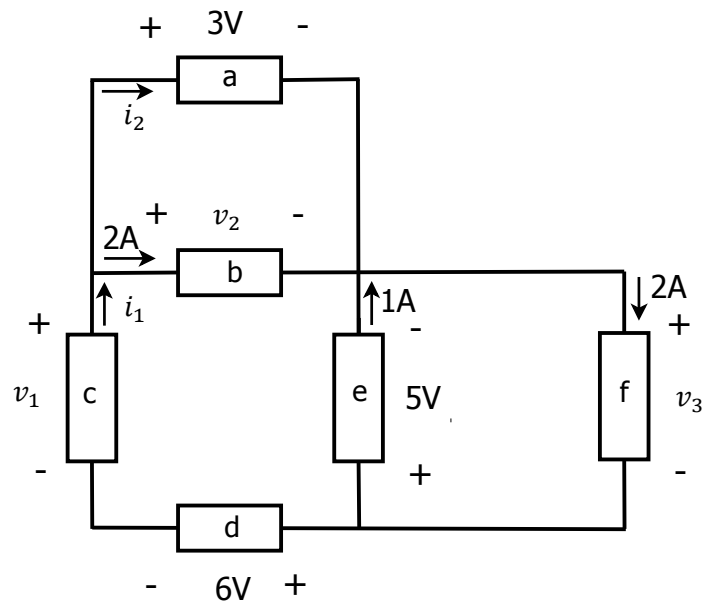
Bereken

- (a)  $v_1$
- (b)  $v_2$
- (c)  $i_1$
- (d) die drywing ge-assosieer met komponent e.

**Question 1.3 [4]**

Calculate

- (a)  $v_1$
- (b)  $v_2$
- (c)  $i_1$
- (d) the power associated with component e.

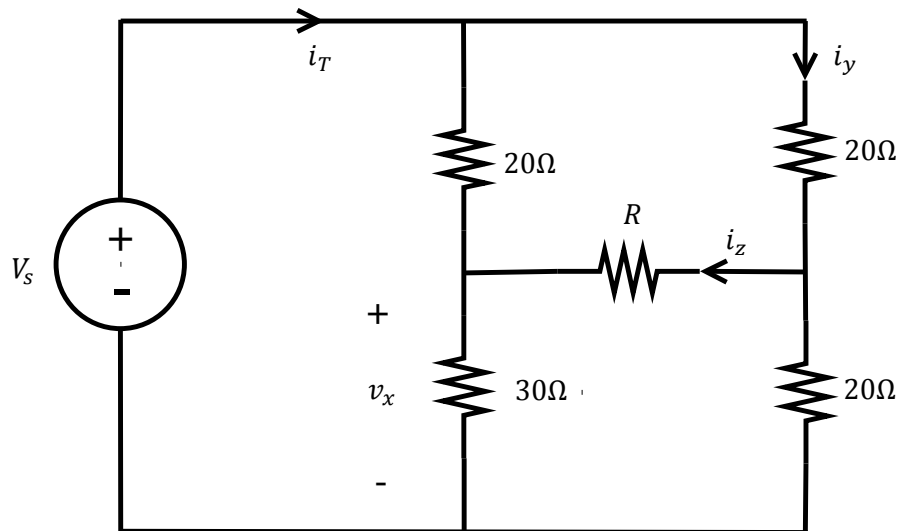


**Vraag 1.4 [8]**

- (a) Aanvaar  $R = 20\ \Omega$ , en bereken die total weerstand soos gesien vanaf die spanningsbron se terminale.  
(b) Aanvaar  $V_s = 20\text{ V}$  en  $R = \infty\ \Omega$ , en bereken  $v_x$ .  
(c) Aanvaar  $i_T = 2\text{ A}$  en  $R = 0\ \Omega$ , en bereken  $i_y$ .  
(d) Aanvaar  $V_s = 22\text{ V}$  en  $R = 0\ \Omega$ , en bereken  $i_z$ .

**Question 1.4 [8]**

- (a) Assume  $R = 20\ \Omega$ , and calculate the total resistance as seen from the terminals of the voltage source.  
(b) Assume  $V_s = 20\text{ V}$  and  $R = \infty\ \Omega$ , and calculate  $v_x$ .  
(c) Assume  $i_T = 2\text{ A}$  and  $R = 0\ \Omega$ , and calculate  $i_y$ .  
(d) Assume  $V_s = 22\text{ V}$  and  $R = 0\ \Omega$ , and calculate  $i_z$ .



Nog spasie vir Vraag 1.4 op die volgende bladsy !!!/More space for Question 1.4 on the following page!!!

Nog spasie vir Vraag 1.4/More space for Question 1.4

## VRAAG/QUESTION 2 [25]

### **Vraag 2.1 [4]**

Gebruik lus-analise om drie vergelykings van die volgende vorm op te stel vir die stroombaan:

$$ai_1 - 20i_2 = b$$

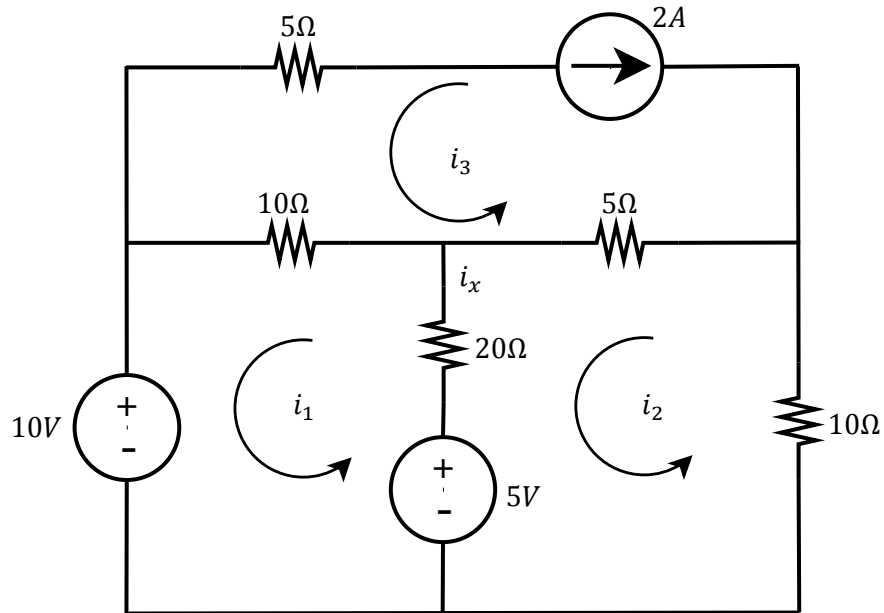
$$-20i_1 + ci_2 = d$$

### **Question 2.1 [4]**

Use mesh-analysis to set up three equations of the following form for the circuit:

$$ai_1 - 20i_2 = b$$

$$-20i_1 + ci_2 = d$$



**Vraag 2.2 [4]**

Gebruik node-analise om two vergelykings van die volgende vorm op te stel vir die stroombaan:

$$3V_1 + eV_2 = f$$

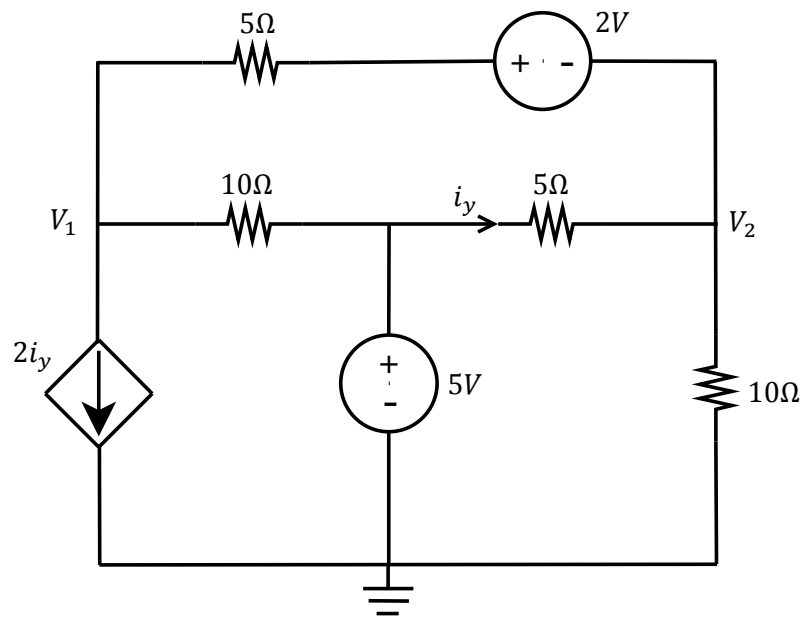
$$gV_1 + hV_2 = 6$$

**Question 2.2 [4]**

Use nodal-analysis to set up two equations of the following form for the circuit:

$$3V_1 + eV_2 = f$$

$$gV_1 + hV_2 = 6$$



**Vraag 2.3 [4]**(a) Bepaal  $V_I$ .

$$\begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -5 \end{bmatrix}$$

(b) Bepaal  $i_3$ .

$$\begin{bmatrix} 2 & 3 & 0 \\ 3 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

**Question 2.3 [4]**(a) Determine  $V_I$ .

$$\begin{bmatrix} 2 & -1 \\ -3 & 2 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -5 \end{bmatrix}$$

(b) Determine  $i_3$ .

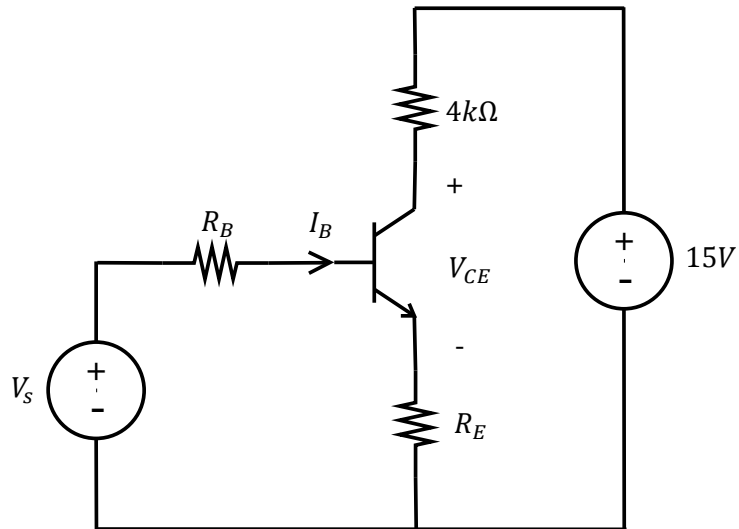
$$\begin{bmatrix} 2 & 3 & 0 \\ 3 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

**Vraag 2.4 [4]**

- (a) Bereken  $V_{CE}$  as  $V_s = 4\text{V}$ ,  $R_E = 0\ \Omega$ ,  $R_B = 100\ \text{k}\Omega$ ,  $\beta = 99$  en  $V_{BE} = 0.7\text{V}$ .
- (b) Bepaal 'n waarde vir  $R_B$  sodat  $V_{CE} = 5\text{V}$ , indien  $V_s = 4\text{V}$ ,  $R_E = 1\ \text{k}\Omega$ ,  $\beta = 99$  en  $V_{BE} = 0.7\text{V}$ .

**Question 2.4 [4]**

- (a) Determine  $V_{CE}$  if  $V_s = 4\text{ V}$ ,  $R_E = 0\ \Omega$ ,  $R_B = 100\ \text{k}\Omega$ ,  $\beta = 99$  and  $V_{BE} = 0.7\text{V}$ .
- (b) Determine a value for  $R_B$  so that  $V_{CE} = 5\text{V}$ , if  $V_s = 4\text{V}$ ,  $R_E = 1\ \text{k}\Omega$ ,  $\beta = 99$  and  $V_{BE} = 0.7\text{V}$ .

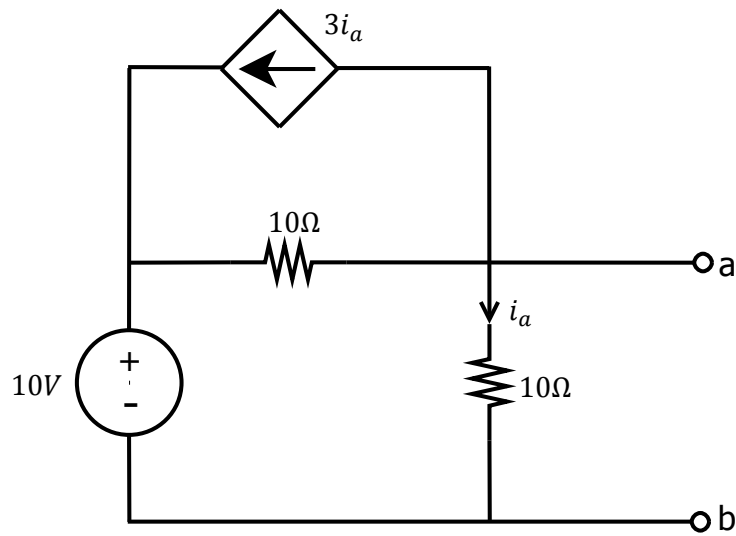


**Vraag 2.5 [4]**

Bepaal die Thevenin-ekwivalente stroombaan ten opsigte van terminale  $a$  en  $b$ .

**Question 2.5 [4]**

Determine the Thevenin equivalent circuit with respect to terminals  $a$  and  $b$ .

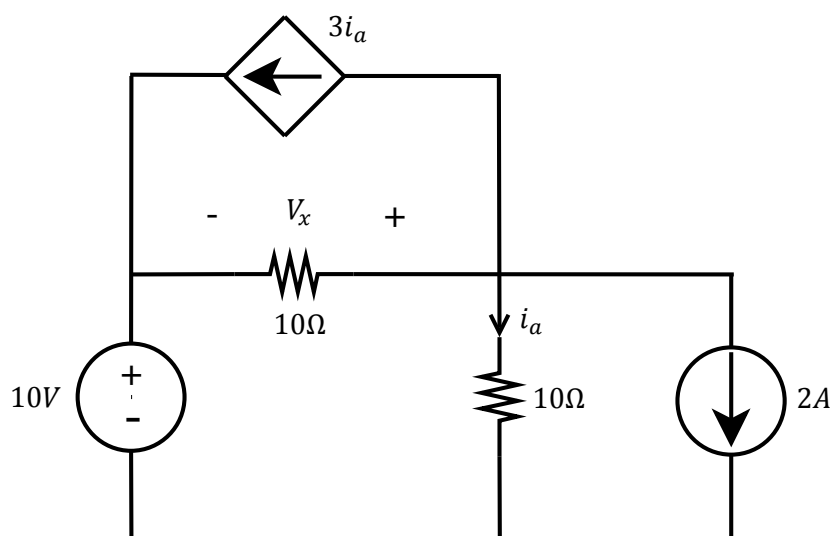


**Vraag 2.6 [4]**

- (a) Bepaal  $V_x$  slegs agv die 10V spanningsbron.  
(b) Bepaal  $V_x$  slegs agv die 2A stroombron.

**Question 2.6 [4]**

- (a) Determine  $V_x$  due to the 10V voltage source only.  
(b) Determine  $V_x$  due to the 2A current source only.

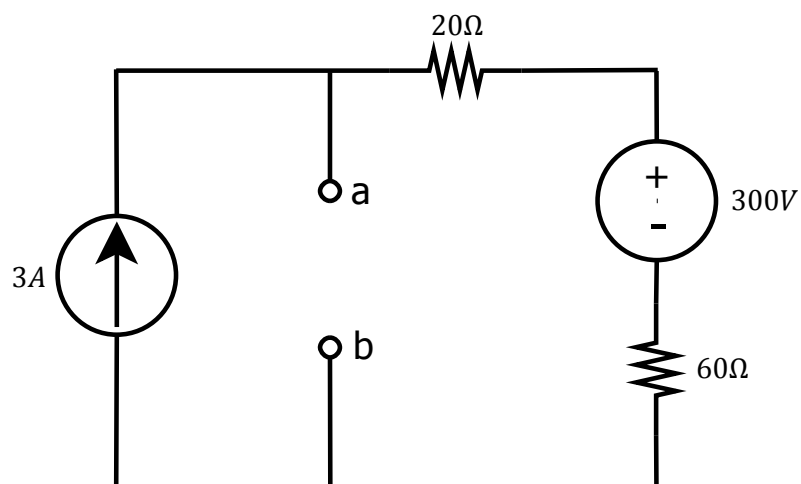


**Vraag 2.7 [2]**

Bepaal die maksimum drywing wat aan 'n optimum weerstand gelewer kan word as dit tussen terminale  $a$  en  $b$  verbind word.

**Question 2.7 [2]**

Determine the maximum power that can be delivered to an optimum resistor when it is connected between terminals  $a$  and  $b$ .



### VRAAG/QUESTION 3 [20]

**Vraag 3.1 [4]**

Gebruik node-analise om twee vergelykings van die volgende vorm op te stel vir die stroombaan:

$$aV_1 + bV_2 = 8$$

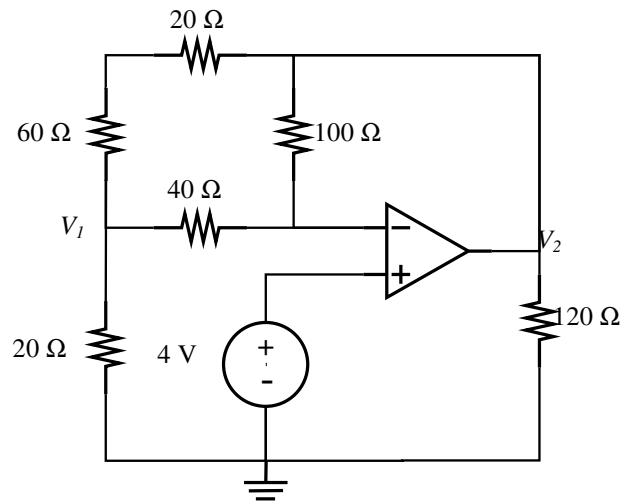
$$cV_1 + dV_2 = 56$$

**Question 3.1 [4]**

Use node analysis to set up two equations of the following form for the circuit:

$$aV_1 + bV_2 = 8$$

$$cV_1 + dV_2 = 56$$



**Vraag 3.2 [4]**

Die bereik van die uitsetspanning  $V_O$  is:

$$-10 \leq V_O \leq -1 \text{ [V]}$$

Die bereik van die insetspanning  $V_S$  is:

$$1 \leq V_S \leq 5 \text{ [V]}$$

Bepaal die minimum en maksimum waarde vir  $R_f$  wat bogenoemde beperkinge bevredig.

**Question 3.2 [4]**

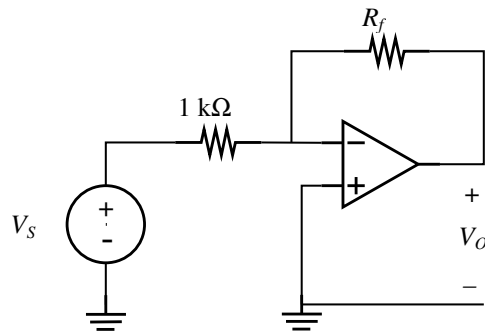
The range of the output voltage  $V_O$  is:

$$-10 \leq V_O \leq -1 \text{ [V]}$$

The range of the input voltage  $V_S$  is:

$$1 \leq V_S \leq 5 \text{ [V]}$$

Determine the minimum and maximum value of  $R_f$  which will satisfy above constraints.

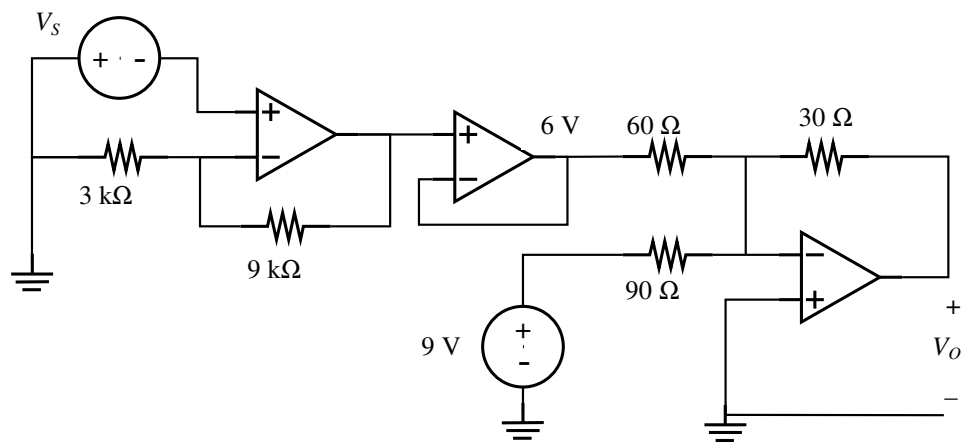


**Vraag 3.3 [4]**

- (a) Bepaal  $V_S$  [2]  
(b) Bepaal  $V_O$  [2]

**Question 3.3 [4]**

- (a) Determine  $V_S$  [2]  
(b) Determine  $V_O$  [2]



**Vraag 3.4 [2]**

Die spanning oor 'n 100 mF kapasitor is  $v(t) = 5 - 5e^{-2t}$  V. Bepaal die kapasitor stroom en drywing vir  $t = 1$  s. (M.a.w. bepaal  $i(1\text{s})$  and  $p(1\text{s})$ .)

**Question 3.4 [2]**

The voltage across a 100 mF capacitor is  $v(t) = 5 - 5e^{-2t}$  V. Determine the capacitor current and power at  $t = 1$  s. (I.e. determine  $i(1\text{s})$  and  $p(1\text{s})$ .)

**Vraag 3.5 [4]**

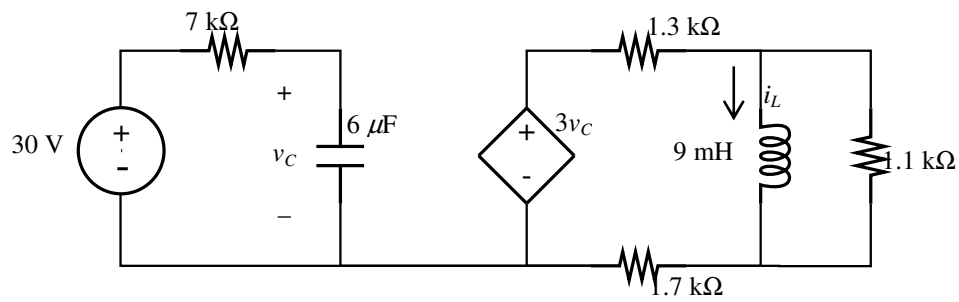
Bepaal:

- a)  $i_L$  en die energie gestoor in die induktor. [2]  
b)  $v_C$  en die energie gestoor in die kapasitor. [2]

**Question 3.5 [4]**

Determine:

- a)  $i_L$  and the energy stored in the inductor. [2]  
b)  $v_C$  and the energy stored in the capacitor. [2]

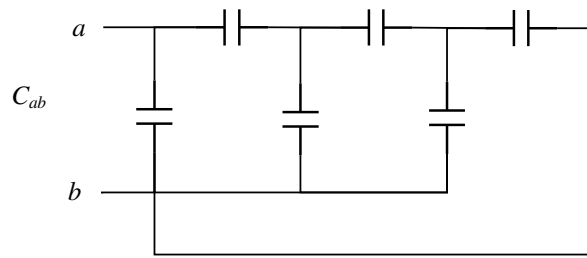


**Vraag 3.6 [2]**

Bepaal die ekwivalente kapasitansie  $C_{ab}$  van die kringbaan. Elke kapasitor het 'n waarde van 6 mF.

**Question 3.6 [2]**

Determine the equivalent capacitance  $C_{ab}$  of the circuit. Every capacitor has a value of 6 mF.



### VRAAG/QUESTION 4 [25]

**Vraag 4.1 [2]**

a) Is  $V_1$  voorlopend of nalopend tot  $V_2$ ? Met hoeveel grade? [1]

b) Is  $V_2$  voorlopend of nalopend tot  $V_3$ ? Met hoeveel grade? [1]

(Skryf die antwoord as volg:

$V_1$  is voorlopend/nalopend tot  $V_2$  met  $\varphi^\circ$ )

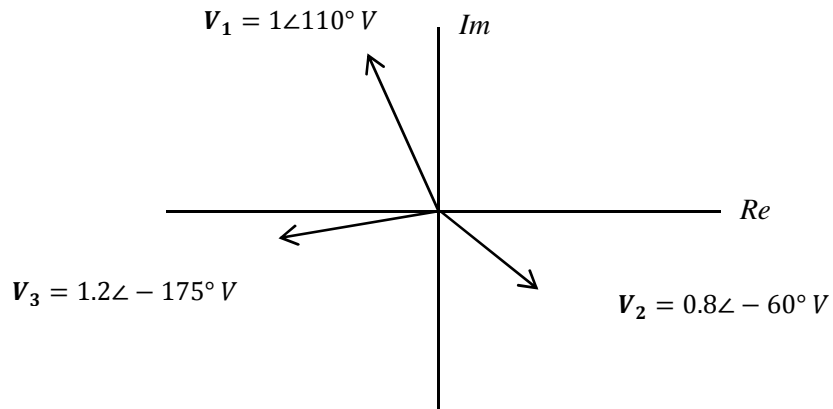
**Question 4.1 [2]**

a) Is  $V_1$  leading or lagging  $V_2$ ? With how many degrees? [1]

b) Is  $V_2$  leading or lagging  $V_3$ ? With how many degrees? [1]

(Write the answer as follows:

$V_1$  is leading/lagging  $V_2$  with  $\varphi^\circ$ )



**Vraag 4.2 [2]**

Gebruik fasors om die volgende as 'n enkele cosinus uitdrukking  $[A\cos(\omega t + \sigma), A > 0, |\sigma| \leq 180^\circ]$  te skryf:  
 $v(t) = -4\cos(\omega t - 240^\circ) + 8\sin(\omega t + 120^\circ)$  V

**Question 4.2 [2]**

Use phasors to write the following as a single cosine expression  $[A\cos(\omega t + \sigma), A > 0, |\sigma| \leq 180^\circ]$ :  
 $v(t) = -4\cos(\omega t - 240^\circ) + 8\sin(\omega t + 120^\circ)$  V

**Vraag 4.3 [4]**

a) Evalueer die volgende en skryf die antwoord in **polêre** vorm:

$$\sqrt{(4 - 3j)^*}$$

b) Evalueer die volgende en skryf die antwoord in **reghoekige** vorm:

$$\frac{-(17e^{-j118.0725^\circ}) + 16 + 30j}{8 + 15j}$$

**Question 4.3 [4]**

a) Evaluate and write the final answer in **polar** form:

$$\sqrt{(4 - 3j)^*}$$

b) Evaluate and write the final answer in **rectangular** form:

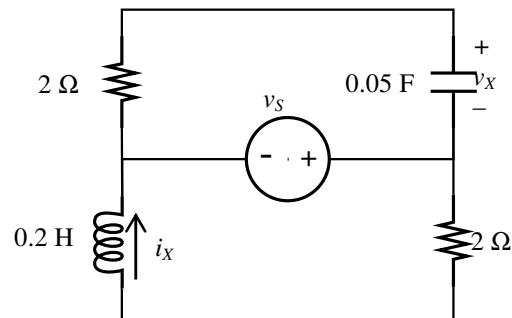
$$\frac{-(17e^{-j118.0725^\circ}) + 16 + 30j}{8 + 15j}$$

**Vraag 4.4 [4]**

Bepaal:

a)  $i_X(t)$  [2]b)  $v_X(t)$  [2](Die finale antwoord **moet** in die volgende vorm wees:[ $A\cos(\omega t + \sigma)$ ,  $A > 0$ ,  $|\sigma| \leq 180^\circ$ ])**Question 4.4 [4]**

Determine:

a)  $i_X(t)$  [2]b)  $v_X(t)$  [2](The final answer **must** be in the following form:[ $A\cos(\omega t + \sigma)$ ,  $A > 0$ ,  $|\sigma| \leq 180^\circ$ ])

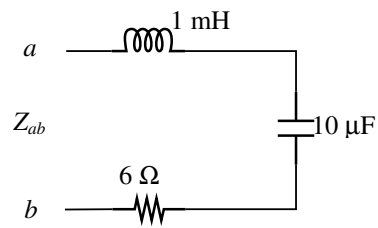
$$v_S(t) = 4\sqrt{2} \cos(10t + 45^\circ)$$

**Vraag 4.5 [2]**

By watter hoekfrekwensie ( $\omega$ ) sal onderstaande kringbaan 'n totale impedansie van  $Z_{ab} = 6 \Omega$  hê?

**Question 4.5 [2]**

At which angular frequency ( $\omega$ ) will the circuit below have a total impedance of  $Z_{ab} = 6 \Omega$ ?



**Vraag 4.6 [4]**

Gebruik node analyse om die komplekse koëffisiënte van die twee onderstaande vergelykings te bepaal:

$$aV_1 + bV_2 = -10 + 40j$$

$$cV_1 + dV_2 = 10$$

Gee die koëffisiënte in **reghoekige** formaat.

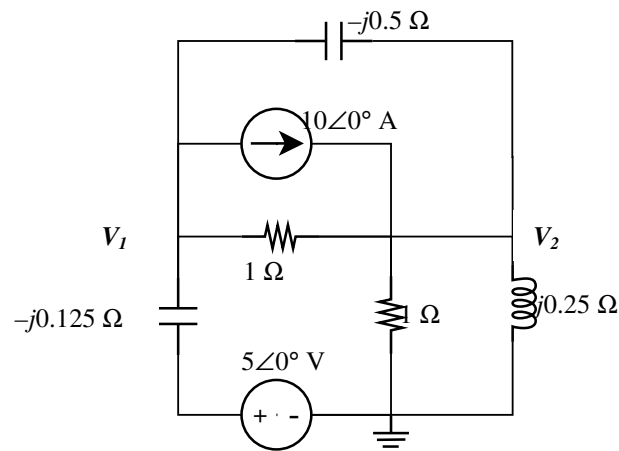
**Question 4.6 [4]**

Use nodal analysis to determine the complex coefficients of the two equations below:

$$aV_1 + bV_2 = -10 + 40j$$

$$cV_1 + dV_2 = 10$$

Give the coefficients in **rectangular** format.



**Vraag 4.7 [4]**

Gebruik lusanalyse om die komplekse koëffisiënte van die twee onderstaande vergelykings te bepaal:

$$eI_1 + fI_2 = 3\angle -150^\circ$$

$$gV_1 + hI_2 = 2\angle 45^\circ$$

Gee die koëffisiënte in **reghoekige** formaat.

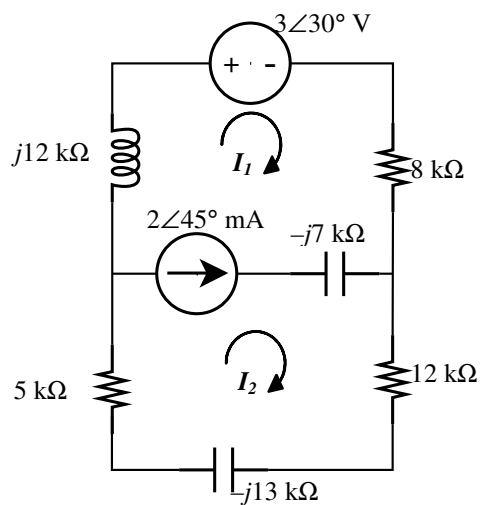
**Question 4.7 [4]**

Use mesh analysis to determine the complex coefficients of the two equations below:

$$eI_1 + fI_2 = 3\angle -150^\circ$$

$$gV_1 + hI_2 = 2\angle 45^\circ$$

Give the coefficients in **rectangular** format.



**Vraag 4.8 [4]**

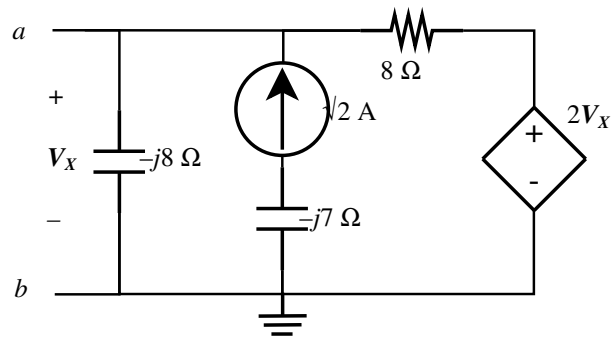
Bepaal die Thevenin ekwivalente spanning en impedansie van die kringbaan ten opsigte van terminale  $a$  en  $b$ .

- a)  $V_{TH}$  in **polêre** formaat.  
b)  $Z_{TH}$  in **reghoekige** formaat.

**Question 4.8 [4]**

Determine the Thevenin equivalent voltage and impedance of the circuit w.r.t. terminals  $a$  and  $b$ .

- a)  $V_{TH}$  in **polar** format.  
b)  $Z_{TH}$  in **rectangular** format.



(Ekstra bladsy vir rofwerk. / *Extra page for rough work.*)

**EBN122 EKSAMEN/EXAMINATION (19 NOVEMBER 2012)****ANTWOORDBLAD/ANSWER SHEET**

|                          |  |                                    |  |  |  |     |     |  |  |
|--------------------------|--|------------------------------------|--|--|--|-----|-----|--|--|
| Van / Surname:           |  | Studentenommer:<br>Student number: |  |  |  |     |     |  |  |
| Voorletters / Initials:  |  | Tel. nr. / Tel no.:                |  |  |  |     |     |  |  |
| Dissipline / Discipline: |  | Taal / Language:                   |  |  |  | Afr | Eng |  |  |

|    |  |    |  |             |            |
|----|--|----|--|-------------|------------|
| Q1 |  | Q2 |  | <b>Tot:</b> | <b>/90</b> |
| Q3 |  | Q4 |  |             |            |

**VRAAG 1 – QUESTION 1 (20)**

|     |            |         |
|-----|------------|---------|
| 1.1 | $p =$      | $W =$   |
| 1.2 | $V =$      | $W =$   |
| 1.3 | $v_I =$    | $v_2 =$ |
|     | $i_I =$    | $p =$   |
| 1.4 | $R_{in} =$ | $v_x =$ |
|     | $i_y =$    | $i_z =$ |

**VRAAG 2 – QUESTION 2 (25)**

|     |                |               |
|-----|----------------|---------------|
| 2.1 | $a =$          | $b =$         |
|     | $c =$          | $d =$         |
| 2.2 | $e =$          | $f =$         |
|     | $g =$          | $h =$         |
| 2.3 | $V_I =$        | $i_3 =$       |
| 2.4 | $V_{CE} =$     | $R_B =$       |
| 2.5 | $V_{TH} =$     | $R_{TH} =$    |
| 2.6 | $V_{x(10V)} =$ | $V_{x(2A)} =$ |
| 2.7 | $P_{max} =$    |               |

| VRAAG 3 – QUESTION 3 (20) |                 |            |
|---------------------------|-----------------|------------|
| 3.1                       | $a =$           | $b =$      |
|                           | $c =$           | $d =$      |
| 3.2                       | $\leq R_f \leq$ |            |
| 3.3                       | $a) V_S =$      | $b) V_O =$ |
| 3.4                       | $i(1s) =$       | $p(1s) =$  |
| 3.5                       | $a) i_L =$      | $w_L =$    |
|                           | $b) v_C =$      | $w_C =$    |
| 3.6                       | $C_{ab} =$      |            |

| VRAAG 4 – QUESTION 4 (25) |                                                   |                                                        |
|---------------------------|---------------------------------------------------|--------------------------------------------------------|
| 4.1                       | $a) V_1$ is _____ $V_2$ met/with _____            |                                                        |
|                           | $b) V_2$ is _____ $V_3$ met/with _____            |                                                        |
| 4.2                       |                                                   |                                                        |
| 4.3                       | $a)$<br><small>(Polêr / Polar)</small>            | $b)$<br><small>(Reghoekig / Rectangular)</small>       |
| 4.4                       | $a) i_X(t) =$                                     | $b) v_X(t) =$                                          |
| 4.5                       | $\omega =$                                        |                                                        |
| 4.6                       | $a =$<br><small>(Reghoekig / Rectangular)</small> | $b =$<br><small>(Reghoekig / Rectangular)</small>      |
|                           | $c =$<br><small>(Reghoekig / Rectangular)</small> | $d =$<br><small>(Reghoekig / Rectangular)</small>      |
| 4.7                       | $e =$<br><small>(Reghoekig / Rectangular)</small> | $f =$<br><small>(Reghoekig / Rectangular)</small>      |
|                           | $g =$<br><small>(Reghoekig / Rectangular)</small> | $h =$<br><small>(Reghoekig / Rectangular)</small>      |
| 4.8                       | $V_{TH} =$<br><small>(Polêr / Polar)</small>      | $Z_{TH} =$<br><small>(Reghoekig / Rectangular)</small> |